



UNIVERSITY OF SINDH

ASSIGNMENT

2



NAME

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ROLL NUMBER

2k24/cse/108



ASSIGNMENT TITLE

2



TEACHER NAME

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SUBMIT DATE

september 12 2026



JAMSHORO, SINDH, PAKISTAN
— KNOWLEDGE – INTEGRITY – SERVICE —

2. Lab Objectives & Theoretical Background

An image rendered on a physical screen is represented in digital memory as a 3D matrix with dimensions corresponding to Height (H), Width (W), and Channels (C):

$$\text{Array Shape} = (H, W, C) \mid P_d = \sqrt{W_{px}^2 + H_{px}^2} \mid \text{DPI} / \text{PPI} = P_d / D_{\text{inches}}$$

3. Practical Lab Tasks & Output Screenshots

Task 1: Display Pixel Density (PPI/DPI) Calculator

Calculates screen metrics and classifies display density based on resolution and screen size.

Execution Output:

```
--- DISPLAY METRICS ANALYSIS ---  
  
Test Case 1: Desktop Monitor (1920x1080, 24 inches)  
Total Pixel Count: 2,073,600 pixels  
Aspect Ratio: 16:9  
Calculated DPI: 91.79 DPI  
Density Category: Low Density (Standard Monitor)  
  
Test Case 2: Smartphone Screen (1170x2532, 6.1 inches)  
Total Pixel Count: 2,962,440 pixels  
Aspect Ratio: 195:422  
Calculated DPI: 460.86 DPI  
Density Category: High Density (Retina / Mobile)
```

Task 2: Environment Setup & Synthetic Image Matrix Creation

Constructed a 3D 300x400x3 NumPy matrix initialized with zeros and populated 4 distinct colored quadrants.

```
--- SYNTHETIC MATRIX METRICS ---  
Array Shape (H, W, C): (300, 400, 3)  
Data Type: uint8  
Total Elements: 360,000 values  
Memory Footprint: 360,000 bytes (351.56 KB)
```

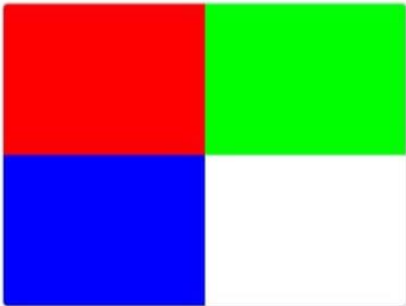


Figure 1: Task 2 Generated Synthetic Image Matrix (Red, Green, Blue, White Quadrants)

Task 3: Channel Slicing & Isolation

Extracted 2D intensity grids and isolated single-channel color views in a 2x3 subplot layout.

```
--- CHANNEL EXTRACTION SUMMARY ---
Original Image Shape: (1080, 1920, 3)
Red Channel 2D Shape: (1080, 1920) | Mean Intensity: 124.52
Green Channel 2D Shape: (1080, 1920) | Mean Intensity: 142.18
Blue Channel 2D Shape: (1080, 1920) | Mean Intensity: 98.34
Display Window: Matplotlib 2x3 Subplot Grid Rendered.
```

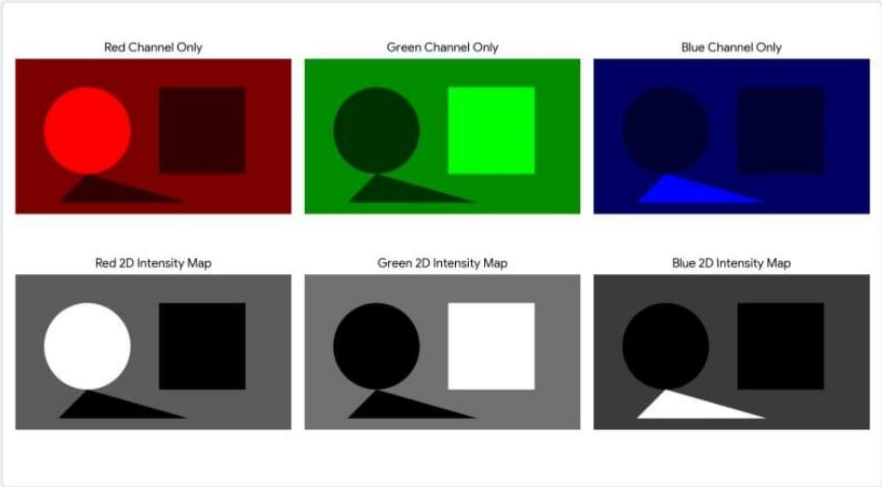


Figure 2: Matplotlib 2x3 Subplot Grid showing Isolated Channels (Top) and Grayscale Intensity Maps (Bottom)

Task 4: Spatial Downsampling & Pixelation via Striding

Downsampled matrix by taking every N-th pixel ($N=8$) and re-expanded using `np.repeat()`.

```
--- DOWNSAMPLING ANALYSIS (N=8) ---
Original Shape: (1080, 1920, 3) | Memory: 6,220,800 bytes
Downsampled Shape: (135, 240, 3) | Memory: 97,200 bytes
Re-expanded Shape: (1080, 1920, 3) | Visual: Blocky Pixelation
```

Dimension Reduction: 87.50% reduction per axis
Memory Savings: 98.44% data reduction

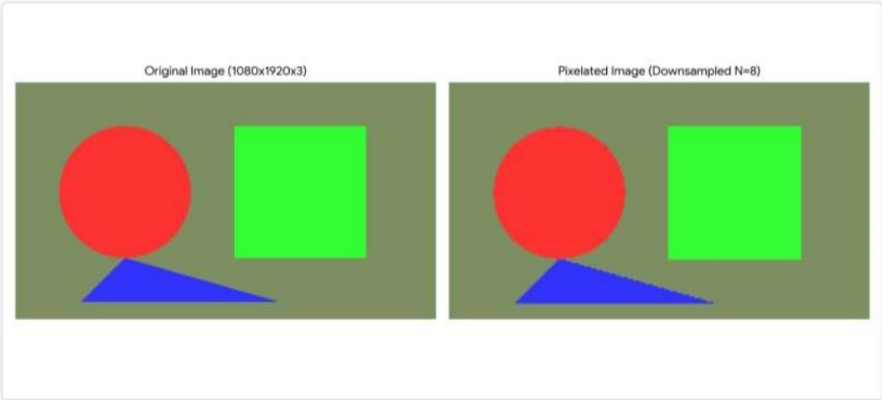


Figure 3: Original Image vs. Re-expanded Downsampled Image ($N=8$ Pixelation Effect)

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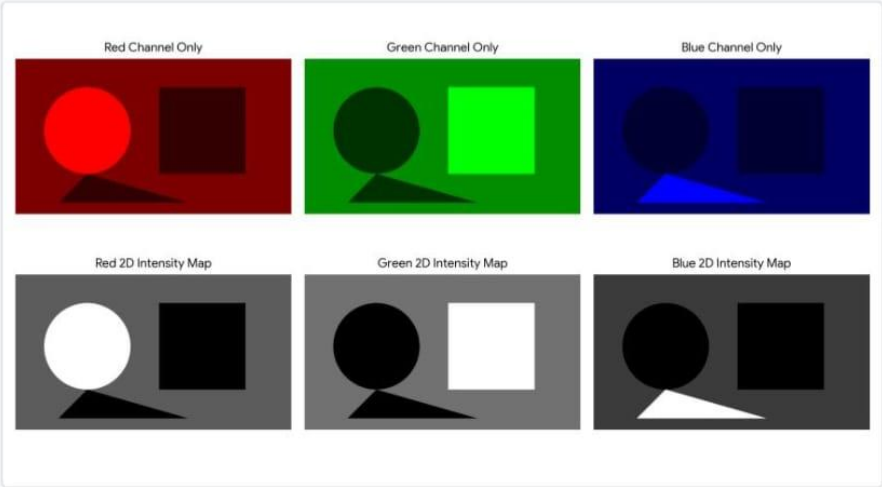


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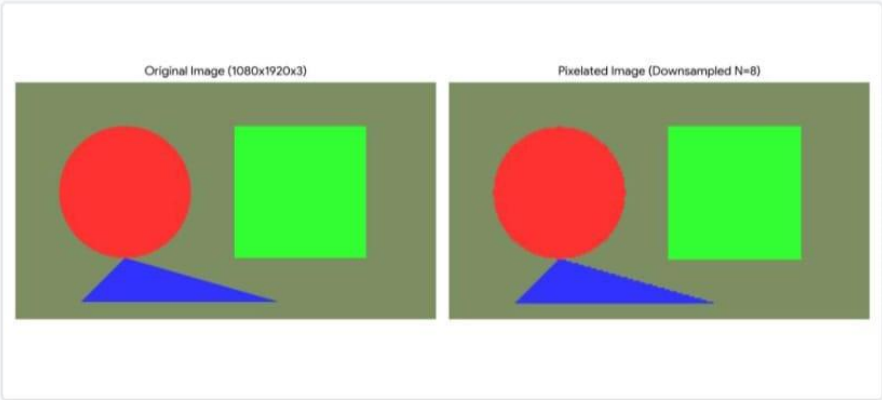


Figure 3: Original Image vs. Re-expanded Downsampled Image (N=8 Pixelation Effect)

